Internet of Things

Final Project

Smart Energy Monitor

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Introduction

The main idea of this project is to design and implement a Smart IoT-Based Energy Monitoring System that can help reduce unnecessary electricity consumption in homes and rooms. The system combines energy monitoring, occupancy detection, and wireless communication to create a smart and efficient solution. The project uses an ESP32 microcontroller as the main processing unit because it provides built-in WiFi functionality and can easily communicate with IoT applications such as Blynk. The ACS712 current sensor is used to measure the current being consumed by the load in real time. Using this current value, the system calculates power consumption and energy usage. These measurements are then sent to the Blynk mobile application, where the user can monitor the readings remotely. A PIR motion sensor is also used to detect whether a person is present in the room or not. An LED is used in place of a room light for demonstration purposes.

The significance of this project is that it promotes energy efficiency and smart automation. Many times, lights and appliances remain switched on even when nobody is present in the room, which leads to unnecessary energy wastage. In this system, when the PIR sensor detects that the room is empty while the light is still ON, a notification is automatically sent to the user through the Blynk app. The user can then remotely turn OFF the light using a button in the mobile application. The Serial Monitor also displays real-time outputs such as current, power, energy usage, room occupancy status, LED status, and alert messages. One new idea used in this project is the integration of IoT-based remote monitoring with occupancy-based energy management. This combination improves user convenience, reduces electricity wastage, and demonstrates the practical application of smart home automation concepts.

Problem Analysis

The problem addressed in this project is the unnecessary consumption of electrical energy in homes and rooms due to lights or appliances being left ON even when no person is present. In many cases, users forget to switch OFF lights before leaving a room, which causes energy wastage and increases electricity costs. Another issue is that most traditional systems do not provide real-time monitoring of electricity usage, so users are often unaware of how much power is being consumed. The project asks us to design a smart IoT-based system that can both monitor energy usage and reduce unnecessary power consumption through automation and remote monitoring.

To solve this problem, the project uses an ESP32 microcontroller as the main controller because it supports both processing and WiFi communication. An ACS712 current sensor is connected to the system to measure the current consumed by the load in real time. Using this value, the ESP32 calculates power and energy consumption. These readings are continuously displayed on the Serial Monitor and are also sent to the Blynk mobile application through WiFi. A PIR motion sensor is used to detect room occupancy. If motion is detected, the system assumes that a person is present in the room. If no motion is detected while the LED light is still ON, the ESP32 sends an alert notification to the user through the Blynk app, informing them that the light should be turned OFF.

The user can remotely control the LED using a button available in the mobile application. This demonstrates how IoT technology can be used for smart energy management and home automation. The system therefore solves both problems by providing real-time energy monitoring and reducing unnecessary electricity usage through occupancy-based control and wireless user interaction.

Design Requirements

The design requirements of this project focus on developing a smart IoT-based energy monitoring and control system that can monitor electricity usage, detect room occupancy, and provide remote control through a mobile application. The system is required to take multiple inputs, process the information, and generate appropriate outputs according to the given conditions. The main inputs of the system are the ACS712 current sensor, the PIR motion sensor, and the commands received from the Blynk mobile application. The ACS712 sensor provides an analog signal proportional to the current consumed by the connected load. This signal is processed by the ESP32 to calculate current, power, and energy consumption in real time. The PIR sensor provides a digital signal indicating whether motion is detected in the room or not. The Blynk application also acts as an input source because the user can remotely send commands to turn the LED ON or OFF.

The outputs of the system include the LED status, mobile notifications, Blynk application readings, and Serial Monitor results. The LED acts as the room light in this project and is controlled using the mobile application. The Blynk app displays current, power, energy usage, and room occupancy status in real time. When no person is detected while the light is still ON, the system sends an alert notification to the user through the application. The Serial Monitor also continuously displays system outputs such as current, power, energy consumption, motion status, LED state, and alert messages for monitoring and debugging purposes.

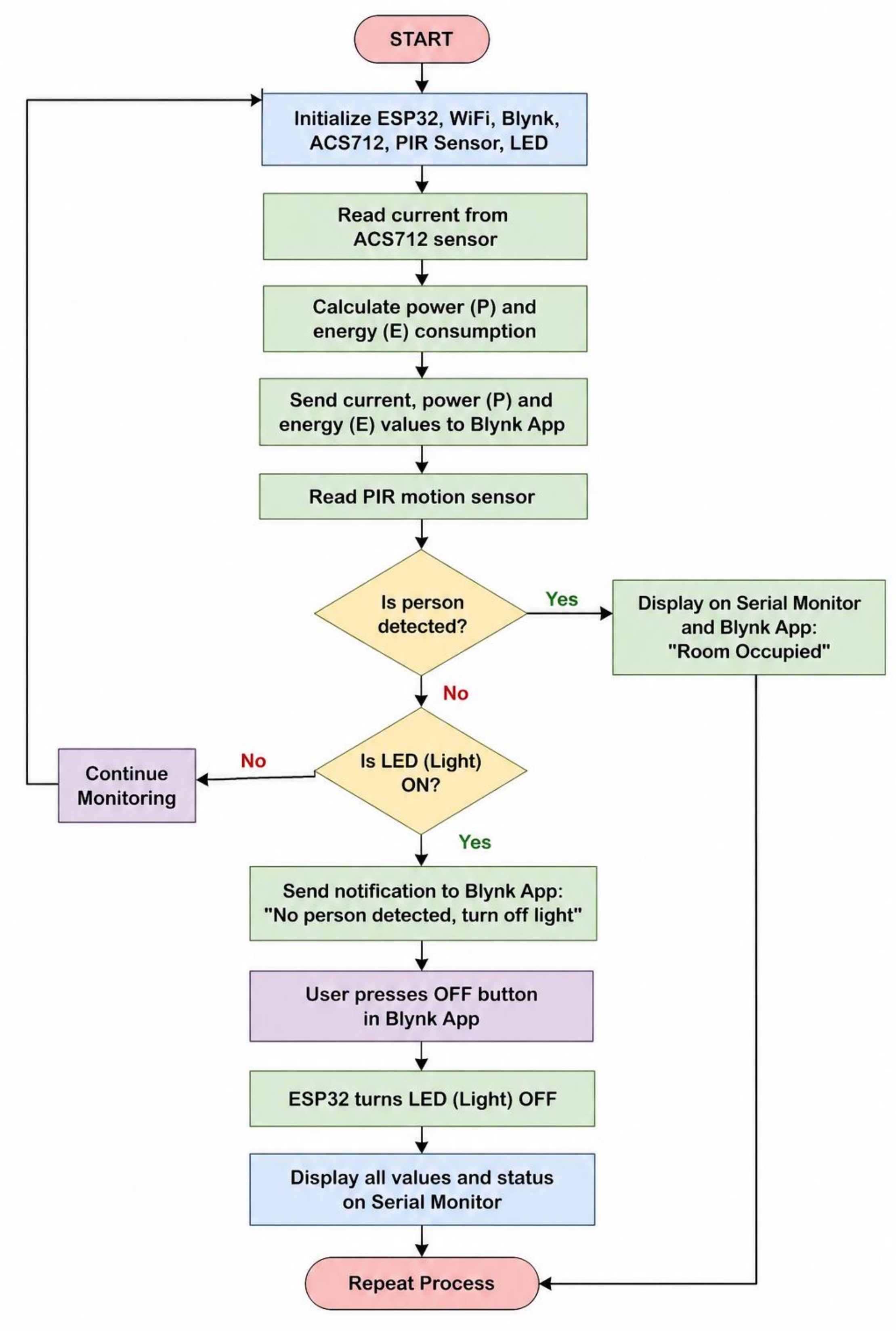
Several constraints are present in the design. The system depends on a stable WiFi connection for communication with the Blynk cloud platform. Sensor readings may contain small fluctuations and noise, especially in current measurements, so filtering and threshold conditions are required. The PIR sensor also has a limited detection range and may occasionally miss small movements. In addition, the system is designed for low-power demonstration purposes using an LED instead of a high-voltage household appliance to ensure safety and simplicity during implementation.

Possible Solutions

There are multiple possible solutions for reducing unnecessary energy consumption in a room. One simple solution is to use a manual switch, where the user turns the light ON and OFF by themselves. This is easy and low-cost, but it depends completely on the user and does not solve the problem if the user forgets to turn the light OFF. Another solution is to use only a PIR motion sensor with automatic light control. In this method, the light turns ON when motion is detected and turns OFF when no motion is detected. This saves energy, but it does not measure electricity usage and does not provide remote monitoring. A third solution is to use only an energy meter with IoT monitoring. This allows the user to see current, power, and energy consumption on a mobile app, but it does not know whether the room is occupied or empty.

The best solution is to combine energy monitoring, occupancy detection, and IoT control in one system. This is the method used in our project. The ESP32 receives current readings from the ACS712 sensor, detects room occupancy using the PIR sensor, and sends all data to the Blynk mobile app. If the room is empty but the LED light is still ON, the system sends a notification to the user. The user can then turn OFF the LED using the app button. This solution is better because it does not only monitor energy usage, but also helps reduce wastage through smart decision-making. Compared to the other methods, it provides real-time measurement, remote access, user notification, and practical energy-saving control. Therefore, the combined IoT-based solution is the most suitable and complete approach for this project.

Preliminary Design



Design Description

The proposed system consists of several important hardware and software blocks that work together to perform smart energy monitoring and control. The main block of the system is the ESP32 microcontroller, which acts as the central processing and communication unit. It receives data from the sensors, performs calculations, communicates with the Blynk cloud server through WiFi, and controls the LED according to user commands and sensor conditions. The ESP32 was selected because it has built-in WiFi support, multiple GPIO pins, and sufficient processing capability for IoT applications.

The ACS712 current sensor block is used for real-time current measurement. The sensor produces an analog output voltage proportional to the current flowing through the load. The ESP32 reads this analog signal through its ADC pin and calculates the current value. Using the measured current and the supply voltage, the system calculates power consumption in watts and energy consumption in kilowatt-hours. These values are displayed on both the Serial Monitor and the Blynk mobile application.

The PIR motion sensor block is responsible for room occupancy detection. It continuously senses infrared radiation changes caused by human movement. When motion is detected, the PIR sensor sends a HIGH signal to the ESP32; otherwise, it sends a LOW signal. This information is used to determine whether the room is occupied or empty.

The LED block acts as the room light or electrical load in this project. The LED can be turned ON or OFF remotely through the Blynk application using a virtual button. If the PIR sensor detects that no person is present while the LED remains ON, the ESP32 sends an alert notification to the user through the Blynk app. The Blynk application block is used for remote monitoring and control. It displays current, power, energy usage, and motion status in real time while also allowing the user to control the LED remotely. Finally, the Serial Monitor block is used for debugging and system monitoring by displaying real-time measurements, occupancy status, LED state, and alert messages during system operation.

Software Simulation

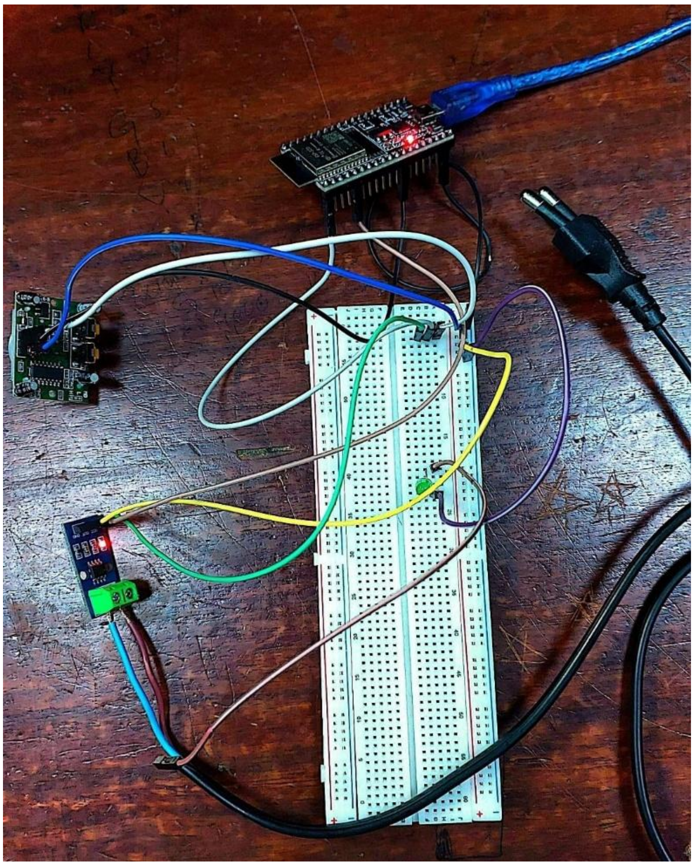
```cpp
#define BLYNK_TEMPLATE_ID   "TMPL6-XY4350L"
#define BLYNK_TEMPLATE_NAME "IOT"
#define BLYNK_AUTH_TOKEN    "aCWuGwtd1LKLHTHk9yIHNnHzrrDcMVvF"
#include <WiFi.h>
#include <BlynkSimpleEsp32.h>
#define CURRENT_SENSOR_PIN  35   // ACS712 OUT
#define PIR_PIN             34   // PIR OUT
#define RELAY_PIN            5   // Relay
float sensitivity   = 0.066;    // 30A module: 66 mV/A
float mainsVoltage  = 220.0;    // Pakistan mains
float current       = 0;
float power         = 0;
float energy_kWh    = 0;
unsigned long lastTime = 0;
int  ledState   = 0;      // tracks Blynk V5 command  (0=OFF,1=ON)
bool alertSent  = false;
char ssid[] = "Zed";
char pass[] = "ghip7604";
BlynkTimer timer;
BLYNK_WRITE(V5) {
  ledState = param.asInt();
  // Relay is active-LOW (same wiring as your working code)
  digitalWrite(RELAY_PIN, ledState == 1 ? LOW : HIGH);
  if (ledState == 0) alertSent = false;
}
void readEnergyData() {
  const int samples = 500;
  long sum = 0;

  for (int i = 0; i < samples; i++) {
    sum += analogRead(CURRENT_SENSOR_PIN);
  }
  float sensorValue   = (float)sum / samples;
  float sensorVoltage = (sensorValue / 4095.0f) * 3.3f;
  float offsetVoltage = 1.65f;
  current = fabsf((sensorVoltage - offsetVoltage) / sensitivity);
  if (current < 0.08f) current = 0;   // noise floor
  power = mainsVoltage * current;
  unsigned long now   = millis();
  float timeHours     = (now - lastTime) / 3600000.0f;
  lastTime            = now;
  energy_kWh         += (power / 1000.0f) * timeHours;
  Blynk.virtualWrite(V1, current);    // A
  Blynk.virtualWrite(V2, power);      // W
  Blynk.virtualWrite(V3, energy_kWh); // kWh
  Serial.printf("Current: %.3f A | Power: %.1f W | Energy: %.4f kWh\n",
                current, power, energy_kWh);
}
void checkRoomAndLight() {
  int motion = digitalRead(PIR_PIN);
  Blynk.virtualWrite(V6, motion);   // 1 = person present
  if (motion == LOW && ledState == 1 && !alertSent) {
    Blynk.logEvent("turn_off_light","No person detected in the room. Please turn off the light.");
    Serial.println("ALERT SENT");
    alertSent = true;
  }
  if (motion == HIGH) alertSent = false;
  Serial.printf("Motion: %d | Relay/LED state: %d\n", motion, ledState);
}
void setup() {
  Serial.begin(115200);
  pinMode(CURRENT_SENSOR_PIN, INPUT);
  pinMode(PIR_PIN, INPUT);
  pinMode(RELAY_PIN, OUTPUT);
  digitalWrite(RELAY_PIN, HIGH);   // relay OFF on boot (active-LOW)
  lastTime = millis();
  Blynk.begin(BLYNK_AUTH_TOKEN, ssid, pass);
  timer.setInterval(2000L, readEnergyData);
  timer.setTimeout (1000L, []() {           // start PIR 1 s later …
    timer.setInterval(2000L, checkRoomAndLight); // … then every 2 s
  });
}
void loop() {
  Blynk.run();
  timer.run();
}
```

Experimental Results

Output:

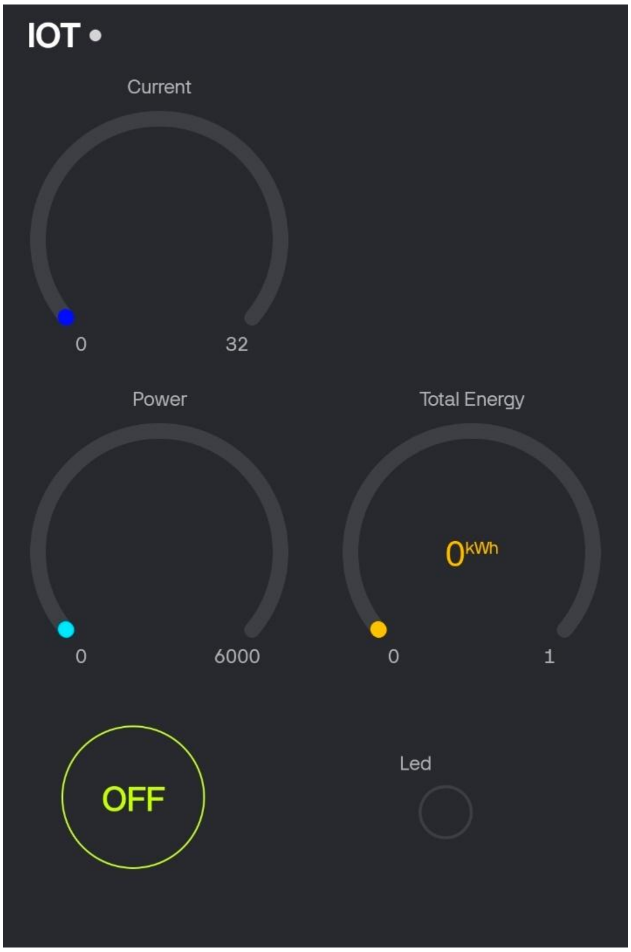
```
[1187] Connecting to Zed
[3521] Connected to WiFi
[3522] IP: 192.168.1.7
[3524] Connecting to blynk.cloud:80
[5879] Ready (ping: 18ms)

Current: 0.000 A | Power: 0.0 W | Energy: 0.0000 kWh
Motion: 1 | Relay/LED state: 0

Current: 0.094 A | Power: 20.7 W | Energy: 0.0000 kWh
Motion: 1 | Relay/LED state: 1

Current: 0.097 A | Power: 21.3 W | Energy: 0.0001 kWh
Motion: 1 | Relay/LED state: 1

Current: 0.096 A | Power: 21.1 W | Energy: 0.0001 kWh
Motion: 0 | Relay/LED state: 1
ALERT SENT
```

Hardware:



Experimental Results

App Interface:



Performance Analysis

The implemented smart IoT-based energy monitoring system successfully achieved the required objectives of real-time energy monitoring, occupancy detection, remote control, and energy-saving notification. During testing, the ACS712 current sensor was able to continuously measure the current consumed by the LED load, and the ESP32 successfully calculated the corresponding power and energy consumption values. These readings were displayed on both the Serial Monitor and the Blynk mobile application in real time. The system showed stable performance during continuous operation, and the calculated values updated correctly after every measurement cycle. The Serial Monitor displayed outputs such as current, power, energy usage, room occupancy status, LED state, and alert notifications, which helped verify proper system operation and simplified debugging.

The PIR motion sensor also performed effectively in detecting room occupancy. When motion was detected, the system identified the room as occupied, and when no motion was detected, the room status changed accordingly. The notification feature worked successfully when the room became empty while the LED remained ON. In this condition, the ESP32 sent an alert notification to the Blynk application, informing the user to turn OFF the light. The LED could then be controlled remotely through the Blynk app button, demonstrating successful IoT-based remote switching and user interaction.

The overall performance of the system was satisfactory for a low-cost smart home energy management solution. The ESP32 maintained stable WiFi communication with the Blynk cloud server and responded quickly to sensor changes and user commands. Small fluctuations were observed in the current sensor readings due to analog sensor noise and ADC variations, but these were minimized using threshold filtering in the code. The PIR sensor also has a limited detection range and may occasionally miss very small movements, but overall occupancy detection remained reliable during testing. The project successfully demonstrated the practical implementation of IoT, sensor interfacing, wireless communication, and smart automation for reducing unnecessary energy consumption.

Conclusion

In conclusion, the developed Smart IoT-Based Energy Monitoring System successfully fulfilled the main objectives of the project by combining energy monitoring, occupancy detection, remote control, and IoT-based communication in a single system. The project demonstrated how modern embedded systems and IoT technologies can be used to reduce unnecessary electricity consumption and improve user convenience. The ESP32 microcontroller successfully interfaced with the ACS712 current sensor, PIR motion sensor, LED load, and Blynk mobile application to create a complete smart monitoring and control solution.

The system was able to measure current, power, and energy consumption in real time and display these values on both the Blynk application and the Serial Monitor. The PIR sensor effectively detected room occupancy, and the notification feature worked properly when the room became empty while the light remained ON. The user was also able to remotely control the LED through the Blynk application, which demonstrated successful wireless communication and remote automation. The Serial Monitor outputs further verified the correct operation of the system by continuously displaying measurements, occupancy status, LED state, and alert messages.

The project also highlighted the importance of energy-efficient smart systems in modern homes and buildings. Although minor sensor noise and PIR detection limitations were observed, the overall system performance remained stable and reliable. The solution is low-cost, easy to implement, and can be expanded further for real household applications by adding multiple appliances, automatic switching, cloud data storage, or advanced energy analytics. Overall, the project provided valuable practical experience in IoT system design, sensor interfacing, wireless communication, embedded programming, and smart automation while successfully addressing the problem of unnecessary energy consumption.

References